

Autonomous AI-Driven Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: The Autoresearch Agentic Framework Approach

Research Report

Autoresearch Project (Claude AI Agents)

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Abstract

We present a systematic computational investigation of the sixty-year-old open problem of whether three mutually orthogonal Latin squares (3-MOLS) of order 10 exist, i.e., whether $N(10) \geq 3$. Our work is distinguished by a novel methodology: an *autonomous agentic research framework* in which Claude AI agents independently design algorithms, write code, launch parallel compute jobs, monitor results, and update research strategy across 30 multi-week sessions without human intervention.

The agents developed and iteratively refined a suite of algorithms: CP-SAT hint-guided cascade optimization (OR-Tools), large neighborhood search (LNS), SA-CT climbing for pair generation, a near-miss pool with warm-start hints, BFS depth-5 intercalate neighborhood analysis, and feasibility attacks with hard energy constraints. These tools were applied to a catalog of 323+ verified MOLS-10 pairs generated by five construction methods.

Our main findings: (1) The minimum total clash energy $\mathcal{E} = cl_{13} + cl_{23}$ cascaded $37 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow 29 \rightarrow \mathbf{26}$ across 30 sessions; the current all-time record is $\mathcal{E} = \mathbf{26}$ ($cl_{13} = 8, cl_{23} = 18$) for pair `iso_tv_21_0` (fingerprint `ee29e34e`), found at Session 26b via 4-worker CP-SAT hint cascade (seed = 1943590465). (2) Structural analysis proves $\mathcal{E} = 26$ is a *strict local minimum*: exhaustive 1-move neighborhood analysis gives row-swap minimum $\mathcal{E} = 43$, col-swap minimum $\mathcal{E} = 42$, single intercalate minimum $\mathcal{E} = 28$; all 281 double-intercalate combinations yield $\mathcal{E} \geq 26$; five independent CP-SAT trials all converge to the *same canonical* L_3 (unique-attractor basin). A 7200-worker-second feasibility attack with hard constraint $\mathcal{E} \leq 25$ finds 0 SAT and 0 INFEASIBLE (all timeout), giving strong empirical evidence that $\mathcal{E} = 26$ is the global minimum for this pair. (3) **Three structurally independent MOLS-10 pairs all reach $\mathcal{E} = 28$** : the secondary cascade of `iso_tv_21_0` ($cl_{13} = 16, cl_{23} = 12$), `mdecomp_279` (fingerprint `e54e791c`, $cl_{13} = 8, cl_{23} = 20$), and `mdecomp_113` (fingerprint `2469c6dc`, $cl_{13} = 17, cl_{23} = 11$), suggesting a structural platform in the energy landscape at $\mathcal{E} = 28$. (4) Every one of the 344 tested MOLS-10 pairs has $CT(L_1, L_2) \leq 2 < 10$, with Glucose3 UNSAT certificates for all 323 pairs (≈ 42 ms each), giving rise to the *CT barrier conjecture*: if all MOLS-10 pairs satisfy $CT \leq 2$, then $N(10) = 2$.

Collectively, this constitutes the strongest computational evidence to date in favor of $N(10) = 2$, while leaving the question formally open.

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1 The $N(10)$ MOLS Problem: Introduction and Challenges

1.1 Latin Squares and Mutual Orthogonality

Definition 1.1 (Latin square). *A Latin square of order n is an $n \times n$ array L with entries from the symbol set $[n] = \{0, 1, \dots, n-1\}$ such that each symbol appears exactly once in every row and exactly once in every column.*

Definition 1.2 (Orthogonal Latin squares). *Two Latin squares L_1 and L_2 of order n are orthogonal, written $L_1 \perp L_2$, if the n^2 ordered pairs $(L_1(i, j), L_2(i, j))$ are all distinct as (i, j) ranges over the n^2 cells. Equivalently, every ordered pair $(a, b) \in [n]^2$ appears in exactly one cell of the superposition.*

Definition 1.3 (MOLS and $N(n)$). *A set $\{L_1, L_2, \dots, L_k\}$ of k Latin squares of order n is called a set of mutually orthogonal Latin squares (MOLS) if $L_i \perp L_j$ for all $i \neq j$. The maximum size k of any such set is denoted $N(n)$.*

When $n = q$ is a prime power, the classical Galois field construction yields $N(q) = q - 1$. For example, $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. In general, the MacNeish bound [4] gives $N(n) \leq n - 1$ for all n .

The question of $N(10)$ is fundamentally harder because $10 = 2 \times 5$ is not a prime power, so no direct algebraic construction applies.

1.2 Historical Background

Euler’s conjecture (1782). Euler [1] conjectured that $N(n) = 1$ (i.e., no two orthogonal Latin squares exist) for all $n \equiv 2 \pmod{4}$, equivalently for $n = 2, 6, 10, 14, 18, \dots$. This conjecture was motivated by the “36 Officers Problem”: can 36 officers of 6 different ranks from 6 different regiments be arranged in a 6×6 square so that no rank or regiment is repeated in any row or column?

Tarry’s verification for $n = 6$ (1901). Tarry [2] proved by exhaustive case analysis that $N(6) = 1$, confirming Euler’s conjecture for $n = 6$. This result — the impossibility of arranging the 36 officers — remains one of the earliest large-scale computational combinatorics results.

Parker’s disproof for $n = 10$ (1960). Parker [3] dramatically disproved Euler’s conjecture by constructing two mutually orthogonal Latin squares of order 10, establishing $N(10) \geq 2$. In the same year, Bose, Shrikhande, and Parker [5] proved that $N(n) \geq 2$ for all $n \neq 2, 6$, completely refuting Euler’s conjecture.

Lam’s nonexistence result (1989). The upper bound $N(10) \leq 8$ comes from a landmark computational result: Lam, Thiel, and Swiercz [6] proved that no projective plane of order 10 exists. Since a projective plane of order n is equivalent to $n - 1$ MOLS of order n , nonexistence of a projective plane of order 10 yields $N(10) \leq 8$. The computation consumed tens of thousands of CPU hours.

Current state. The best known bounds are:

$$2 \leq N(10) \leq 8.$$

The problem of determining $N(10)$ precisely has been open for over 60 years. Most combinatorialists conjecture $N(10) = 2$ [7, 8], but no proof of this conjecture exists. In particular, the existence of a third MOLS of order 10—whether $N(10) \geq 3$ —is completely unknown.

1.3 Mathematical Framework

1.3.1 The Clash Count and Energy Function

Definition 1.4 (Clash count). *For Latin squares A and B of order n , the clash count is*

$$\text{cl}(A, B) = n^2 - |\{(A(i, j), B(i, j)) : i, j \in [n]\}|,$$

i.e., the number of ordered pairs $(a, b) \in [n]^2$ that do not appear in the superposition of A and B . We have $\text{cl}(A, B) = 0$ if and only if $A \perp B$.

For a triple (L_1, L_2, L_3) with $L_1 \perp L_2$ (so $\text{cl}_{12} = 0$), we define the *energy*:

$$\mathcal{E}(L_1, L_2, L_3) = \text{cl}(L_1, L_3) + \text{cl}(L_2, L_3) = \text{cl}_{13} + \text{cl}_{23}.$$

The **goal** is to find (L_1, L_2, L_3) with $\mathcal{E} = 0$, which would prove $N(10) \geq 3$.

1.3.2 Common Transversals

Definition 1.5 (Common transversal). *Let L_1, L_2 be orthogonal Latin squares of order n . A common transversal (CT) of the pair (L_1, L_2) is a set $T = \{(r_0, c_0), \dots, (r_{n-1}, c_{n-1})\}$ of n cells satisfying:*

- (i) *each row index $0, \dots, n-1$ appears exactly once;*
- (ii) *each column index $0, \dots, n-1$ appears exactly once;*
- (iii) *the multiset $\{L_1(r_i, c_i)\}_{i=0}^{n-1}$ consists of all n distinct symbols;*
- (iv) *the multiset $\{L_2(r_i, c_i)\}_{i=0}^{n-1}$ consists of all n distinct symbols.*

We write $\text{CT}(L_1, L_2)$ for the maximum number of pairwise disjoint common transversals of (L_1, L_2) .

The fundamental connection between common transversals and MOLS is:

Theorem 1.6 (CT Necessity). *A Latin square L_3 of order n satisfies $L_3 \perp L_1$ and $L_3 \perp L_2$ if and only if the n^2 cells can be partitioned into n pairwise disjoint common transversals $T_0, \dots, T_{n-1}$ of (L_1, L_2) , where $T_k = \{(i, j) : L_3(i, j) = k\}$.*

Proof sketch. Each symbol k of L_3 occupies exactly one cell per row and one per column (Latin square property). If $L_3 \perp L_1$, those n cells contain each L_1 -symbol exactly once; if $L_3 \perp L_2$, they contain each L_2 -symbol exactly once. Hence the n cells for symbol k form a common transversal T_k . The n transversals $T_0, \dots, T_{n-1}$ partition all n^2 cells. The converse is direct. $\square$

Corollary 1.7 (CT Barrier). *If $\text{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. In particular, for $n = 10$: if $\text{CT}(L_1, L_2) \leq 9$, the pair (L_1, L_2) cannot be extended to three MOLS.*

1.3.3 SAT Encoding

The existence of $L_3 \perp L_1$ and $L_3 \perp L_2$ for fixed (L_1, L_2) is encoded as a SAT instance with 1000 Boolean variables $x_{i,j,k} = 1 \iff L_3(i, j) = k$, plus five families of exactly-one constraints enforcing the Latin square property of L_3 and orthogonality with both L_1 and L_2 . An UNSAT certificate from a CDCL solver (Glucose3 [10]) definitively proves no L_3 exists for that pair.

1.4 Key Challenges

The $N(10)$ problem poses a combination of mathematical and computational challenges that have stymied researchers for over six decades:

1. **Exponential search space.** The number of distinct Latin squares of order 10 is approximately 10^{38} [16]. Even restricting to valid L_3 given fixed (L_1, L_2) , the search space is vast.
2. **Local minima barriers.** Simulated annealing over the energy $\mathcal{E}$ reaches a floor of $\mathcal{E} = 37$ that cannot be escaped by standard local moves (row/col swaps, intercalates). This floor is a strict local minimum under all neighborhood moves tested prior to the introduction of CP-SAT optimization.
3. **The CT barrier.** All 344 MOLS-10 pairs tested in this project have $\text{CT}(L_1, L_2) \leq 2$, far below the required threshold of $n = 10$. By Corollary 1.7, none of these pairs can yield 3-MOLS. Whether *any* MOLS-10 pair can achieve $\text{CT} \geq 10$ is unknown.
4. **No projective plane.** The nonexistence of a projective plane of order 10 rules out the construction of all $n - 1 = 9$ MOLS simultaneously, but gives no direct obstacle to finding 3 MOLS.
5. **Absence of algebraic structure.** Since 10 is not a prime power, there is no Galois field $\text{GF}(10)$, and the standard algebraic MOLS construction does not apply. All known approaches rely on combinatorial search.
6. **Scale of prior computation.** Lam et al. [6] required tens of thousands of CPU hours to settle the related question of the projective plane. A direct exhaustive search for $N(10) \geq 3$ appears computationally infeasible with current resources without major algorithmic advances.

2 The Autoresearch Agentic Framework

2.1 Framework Overview

The *Autoresearch Agentic Framework* is an autonomous AI research system in which Claude-powered agents conduct end-to-end computational research without human intervention. Over 30 sessions spanning multiple weeks, agents executed the following research cycle repeatedly:

1. **State assessment.** Read prior session logs, check the near-miss pool, and summarize current best energies and which pairs have been explored.
2. **Strategy decision.** Based on the current state, decide which algorithms to run, which pairs to target, and what compute budget to allocate.
3. **Implementation.** Write or modify Python/shell code for new algorithms, fix bugs identified in prior sessions, and configure parameters.
4. **Execution.** Launch parallel worker processes, monitor output, kill stalled workers, and save breakthroughs to the near-miss pool.
5. **Analysis.** Interpret results, update the research direction, and document findings for the next session.

This cycle ran in a **15-minute autonomous loop**: within each session, agents continuously checked progress and acted without waiting for human feedback.

2.2 Key Agent Capabilities

The framework enabled capabilities that would be impractical for human researchers working alone:

- **Parallel process management.** Agents spawned dozens of parallel search processes (each using 2–4 CPU workers), monitored their output in real time, killed processes that stalled at local minima, and restarted them with new configurations.
- **Adaptive exploration.** Agents tracked which of the 323+ MOLS-10 pairs had been explored and at what energy, then allocated compute preferentially to the most promising pairs.
- **Breakthrough detection.** When a new energy record was found, the agent immediately updated the near-miss pool, used the new L_3 as a hint for subsequent CP-SAT runs (the hint cascade), and shifted other workers to explore from the new record.
- **Algorithm invention.** Key algorithmic innovations — the hint cascade strategy, the BFS depth-5 local minimum proof, the 4-row LNS analysis, and the feasibility attack — were all conceived and implemented by agents during the course of the research.
- **Cross-session continuity.** Agents persisted state (near-miss pool, pair catalog, best-energy files) across sessions, enabling long-running cascades to span multiple autonomous sessions.

2.3 Algorithm Portfolio

Agents developed and deployed the following suite of algorithms:

2.3.1 CP-SAT Hint-Guided Optimization

The core optimization engine uses Google OR-Tools CP-SAT [9], a constraint-programming solver with integrated SAT and linear programming. For fixed (L_1, L_2) from the near-miss pool, the model minimizes $\mathcal{E} = \text{cl}_{13} + \text{cl}_{23}$ subject to L_3 being a valid Latin square orthogonal to neither but close to both.

Key design choices:

- Decision variables: 100 integer variables $L_3[i, j] \in [10]$.
- Orthogonality coverage indicators: 200 Boolean variables, one per ordered pair (a, b) for each of cl_{13} and cl_{23} .
- Symmetry breaking: fixing L_3 's first row to the identity reduces search by factor $10!$ without loss of generality.
- Parallel workers: 2–4 CP-SAT workers running portfolios of different heuristics.

2.3.2 Hint Cascade Strategy

The agents' most impactful innovation is the *hint cascade*: a chain of CP-SAT runs where each run receives the best L_3 from the previous run as a warm-start hint.

The cascade allows CP-SAT to leverage its non-local branching ability at each step while being guided toward regions already known to be good.

Algorithm 1 Hint Cascade for Pair (L_1, L_2)

Require: MOLS pair (L_1, L_2) with $\text{cl}_{12} = 0$, timeout T , max rounds K

Ensure: Best L_3 and energy $\mathcal{E}^*$ found

```
1:  $L_3^{(0)} \leftarrow$  random Latin square;  $\mathcal{E}^{(0)} \leftarrow \mathcal{E}(L_1, L_2, L_3^{(0)})$ 
2:  $E^* \leftarrow \mathcal{E}^{(0)}$ ;  $L_3^* \leftarrow L_3^{(0)}$ 
3: for  $k = 1, 2, \dots, K$  do
4:   Run CP-SAT with hint  $L_3^{(k-1)}$ , minimize  $\mathcal{E}$ , timeout  $T$ 
5:   Obtain  $(L_3^{(k)}, \mathcal{E}^{(k)})$ 
6:   if  $\mathcal{E}^{(k)} < E^*$  then
7:      $E^* \leftarrow \mathcal{E}^{(k)}$ ;  $L_3^* \leftarrow L_3^{(k)}$ 
8:   end if
9:   if  $\mathcal{E}^{(k)} = 0$  then return  $L_3^*, E^*$  ▷ 3-MOLS found!
10:  end if
11:  if  $\mathcal{E}^{(k)} \geq \mathcal{E}^{(k-1)}$  then break ▷ No improvement
12:  end if
13: end for
14: return  $L_3^*, E^*$ 
```

2.3.3 MOLS Pair Generation Methods

Five methods were used to generate the catalog of 323+ verified MOLS-10 pairs:

1. **Transversal Decomposition (TV).** Fix Parker’s TV-21 square and enumerate its transversals; partition cells into 10 disjoint transversals via Dancing Links [12] to obtain orthogonal mates.
2. **Isotopy Variants (ISO).** Apply row permutations, column permutations, and symbol relabelings to known MOLS pairs to generate new pairs in different isotopy classes.
3. **SA-CT Climbing.** Run simulated annealing [11] with the objective of maximizing $\text{CT}(L_1, L_2)$ while maintaining $\text{cl}_{12} = 0$.
4. **Modular Decomposition (mdecomp).** Construct MOLS pairs from modular arithmetic and transversal decomposition of random Latin squares; generates pairs in novel L_1 -families not isotopic to Parker’s original.
5. **Intercalate Chain (IC).** Apply bounded sequences of intercalate flips to known pairs to generate nearby pairs.

2.3.4 Additional Search Methods

- **SA warm-starts.** Simulated annealing initialized from a near-miss pool entry, allowing the energy to be pushed below the SA floor of $\mathcal{E} = 37$.
- **Large Neighborhood Search (LNS).** Fix a subset of k rows (or columns) of L_3 as free variables; re-optimize using CP-SAT. Agents tested all $\binom{10}{4} = 210$ four-row subsets to prove 4-row local optimality.
- **BFS Intercalate Analysis.** Breadth-first search over all intercalate flips from a given L_3 , up to depth d , to certify that no sequence of d intercalate moves can reduce $\mathcal{E}$.
- **Feasibility Attack.** Run CP-SAT with the hard constraint $\mathcal{E} \leq E_{\text{target}}$ (rather than minimizing $\mathcal{E}$). Returns SAT (found better solution), UNSAT (proved lower bound $\mathcal{E} > E_{\text{target}}$), or TIMEOUT.

- **Glucose3 SAT Certificates.** For each MOLS-10 pair, encode the existence of L_3 as a CNF instance and run Glucose3 [10]. An UNSAT result certifies $\text{CT}(L_1, L_2) < 10$.

2.4 The Near-Miss Pool

All triples (L_1, L_2, L_3) with energy $\mathcal{E} \leq 37$ are stored in a persistent near-miss pool. The pool serves two roles:

1. **Hint source.** The best pool entries provide L_3 hints for CP-SAT cascades.
2. **Progress tracker.** The minimum pool energy is the global benchmark; agents report it at the start of each session.

File-locking (`fcntl.LOCK_EX`) ensures race-condition-free writes when multiple parallel workers update the pool simultaneously.

3 Stage-by-Stage Agent Results

3.1 Overview: The Energy Cascade

Table 1 summarizes the energy cascade achieved across all 30 sessions.

Sessions	Key Development	Best $\mathcal{E}$	Pair
1–8	SA floor established; 323+ pairs generated	37	various
9	CP-SAT breaks SA floor (2 independent pairs)	34	various
10–11	SA warm-start cascade: $34 \rightarrow 33 \rightarrow 32 \rightarrow 31$	31	various
12–14	CP-SAT cascade: $31 \rightarrow 30 \rightarrow 29$	29	4f04a773 class
14b	2-intercalate neighborhood: $30 \rightarrow 29$ instantly	29	4f04a773 class
15–16	4 isotopy classes reach $E=29$; BFS depth-5 proofs	29	4 classes
17–25	Full 323-pair scan; <code>iso_tv_21.0</code> scan record $E=32$	29	ee29e34e
25b	<code>iso_tv_21.0</code> v3 trial 1: cold $E=30$ (new record)	29	ee29e34e
26	<code>iso_tv_21.0</code> v3 trial 2: hint cascade $E=29$	29	ee29e34e
26b	<code>iso_tv_21.0</code> v3 trial 4: $E=26$ ALL-TIME RECORD	26	ee29e34e
27–28	$E=26$ structural analysis; feasibility attack	26	ee29e34e
29–30	<code>mdecomp_279</code> and <code>mdecomp_113</code> reach $E=28$	26	multiple

3.2 Stage 1 (Sessions 1–8): Infrastructure and Pair Generation

3.2.1 Pair Catalog Construction

Agents constructed a catalog of 323 verified MOLS-10 pairs using the five generation methods described in Section 2. Table 2 shows the breakdown by method.

Method	Pairs	CT=1	CT=2	Notes
TV variants (Parker 1960)	56	46	10	Based on TV-21 square
Isotopy transforms (ISO)	22	8	14	Permuted TV-21 variants
Intercalate chains	6	3	3	IC depth 2–12
<code>mdecomp</code>	136	2	134	Novel L_1 families
SA-CT climbing	103	0	103	Random-start CT optimization
Total	323	58	265	All with $\text{cl}_{12} = 0$

3.2.2 The CT Barrier: First Evidence

For all 323 pairs, Glucose3 UNSAT certificates confirmed $\text{CT}(L_1, L_2) \leq 2$. By Corollary 1.7, none of these pairs can be extended to 3-MOLS.

Table 3: CT distribution across 323 MOLS-10 pairs.

CT value	Pairs	Fraction
1	59	18.3%
2	264	81.7%
≥ 3	0	0%

3.2.3 Simulated Annealing Floor: $\mathcal{E} = 37$

A 7-replica parallel-tempering SA [11] searched for L_3 over all pairs. After hundreds of 97-second trials across 8 sessions, the SA floor was firmly established at $\mathcal{E} = 37$. This is a *strict local minimum* under all SA move types (row swap, column swap, symbol relabel, intercalate, k -scramble, big-scramble): no move from any $\mathcal{E} = 37$ state leads to $\mathcal{E} < 37$.

The $\mathcal{E} = 37$ floor held across all pools of starting states, all pairs, and all random seeds. This indicated that breaking the floor required a non-local optimization method.

3.3 Stage 2 (Sessions 9–16): First Cascades, $\mathcal{E} = 37 \rightarrow 29$

3.3.1 Session 9: CP-SAT Breaks the SA Floor

The introduction of OR-Tools CP-SAT in Session 9 immediately broke the SA floor:

Table 4: Session 9 CP-SAT results: breaking $\mathcal{E} = 37$.

Run	cl ₁₃	cl ₂₃	$\mathcal{E}$	Time	Workers	Notes
No symbreak, trial 2	18	18	36	180 s	2	First CP-SAT improvement
Symbreak + hint($\mathcal{E}=39$)	17	17	34	300 s	4	Two independent pairs
Second pair	18	16	34	120 s	2	Independently confirmed

Why CP-SAT succeeds where SA fails. SA moves change one or a few cells at a time; from $\mathcal{E} = 37$, every such move increases energy. CP-SAT’s branch-and-bound with constraint propagation can make large, non-local assignments. With symmetry breaking reducing redundancy by a factor of $10! \approx 3.6 \times 10^6$, CP-SAT navigates directly to regions unreachable by local SA moves.

The fact that two independent (L_1, L_2) pairs both reached $\mathcal{E} = 34$ within the same session ruled out pair-specific luck.

3.3.2 Sessions 10–14: Cascade $\mathcal{E} = 34 \rightarrow 29$

Armed with $\mathcal{E} = 34$ pool entries, SA warm-starts immediately descended further. The agents then alternated SA warm-starts (fast descent) with CP-SAT hint runs (deeper optimization) in a cascade:

- Session 10: SA warm-start from $\mathcal{E} = 34$ pool $\rightarrow \mathcal{E} = 33$ (cl₁₃ = 14, cl₂₃ = 19). File-locking race condition identified and fixed.
- Session 10–11: CP-SAT cascade: $\mathcal{E} = 33 \rightarrow 32 \rightarrow 31$.
- Session 12–13: Further CP-SAT optimization: $\mathcal{E} = 31 \rightarrow 30$.

- Session 14: CP-SAT proof worker finds $\mathcal{E} = 30 \rightarrow 29$ in 311 seconds.
- Session 14b: A 2-intercalate neighborhood search from the $\mathcal{E} = 30$ state *immediately* found $\mathcal{E} = 29$ ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 14$), proving the intercalate neighborhood of $\mathcal{E} = 30$ contains $\mathcal{E} = 29$ states.

3.3.3 Session 16: Four Isotopy Classes Proved at $\mathcal{E} = 29$

By Session 16, four distinct isotopy classes had reached $\mathcal{E} = 29$:

Table 5: The four isotopy classes at $\mathcal{E} = 29$ (Session 16).

Class fingerprint	cl_{13}	cl_{23}	IC count	BFS depth proved
4f04a773	15	14	28	5 (22M+ states)
2cc8e22f	14	15	23	5 (22M+ states)
74bb7b86	17	12	19	5 (22M+ states)
0968fc79	17	12	24	5 (22M+ states)

The BFS depth-5 analysis (over 22 million states each) proved that no sequence of up to 5 intercalate flips from any $\mathcal{E} = 29$ solution leads to $\mathcal{E} < 29$. These are *global intercalate-neighborhood minima at depth 5*.

Structural observation (classes 74bb7b86 and 0968fc79). These two classes share the same L_1 matrix (Hamming distance 0/100) but differ in L_2 (Hamming distance 50/100). Remarkably, all four cross-combinations (L_2, L_3) yield exactly $\mathcal{E} = 29$ with identical clash distributions, revealing a deep structural symmetry in the energy landscape.

3.4 Stage 3 (Sessions 17–26b): Full Scan and $\mathcal{E} = 26$ Record

3.4.1 Complete 323-Pair Scan

Sessions 17–25 performed a systematic cold-start scan of all 323 MOLS-10 pairs, allocating 120–180 seconds of CP-SAT optimization per pair. Key findings:

- 17 “standout” pairs reached $\mathcal{E} \leq 37$ in cold scan.
- iso_tv_21_0 (fingerprint **ee29e34e**, L_1 family #12) achieved the scan record $\mathcal{E} = 32$ ($\text{cl}_{13} = 13$, $\text{cl}_{23} = 19$), making it the top candidate for deep optimization.
- mdecomp_279 (fingerprint **e54e791c**) achieved co-scan-record $\mathcal{E} = 33$ ($\text{cl}_{13} = 12$, $\text{cl}_{23} = 21$), the first standout from a non-TV L_1 family.
- 28 distinct L_1 families were identified across all 323 pairs.
- The scan completed at Session 26b: all 323/323 pairs processed.

3.4.2 iso_tv_21_0: The Hint Cascade to $\mathcal{E} = 26$

Agents focused the hint cascade on iso_tv_21_0 (fingerprint **ee29e34e**), launching a dedicated “v3” push with 4 parallel CP-SAT workers:

Table 6: iso_tv_21_0 v3 hint cascade achieving the $\mathcal{E}=26$ all-time record.

Trial	Hint	cl_{13}	cl_{23}	$\mathcal{E}$	Notes
1 (cold)	none	15	15	30	New push co-record!
2	E=30 hint	14	15	29	seed=1333021984
3	E=29 hint	14	15	29	No improvement
4	E=29 hint	8	18	26	seed=1943590465, ALL-TIME RECORD

The E=26 result ($cl_{13} = 8$, $cl_{23} = 18$, seed=1943590465): the cascade jumped from $\mathcal{E} = 29$ to $\mathcal{E} = 26$ in a single 4-worker CP-SAT trial. The trajectory $32 \rightarrow 30 \rightarrow 29 \rightarrow 26$ required only four trials total.

This is a drop of 3 energy units in a single step, demonstrating CP-SAT’s ability to make large non-local jumps in the energy landscape that are inaccessible to intercalate- based local search.

3.5 Stage 4 (Sessions 27–28): Structural Analysis of $\mathcal{E} = 26$

3.5.1 Proving $\mathcal{E} = 26$ is a Strict Local Minimum

Agents performed an exhaustive analysis of the 1-move neighborhood of the $\mathcal{E} = 26$ state:

Table 7: Exhaustive 1-move neighborhood of the E=26 state.

Move type	Neighborhood size	Min $\mathcal{E}$ observed	Result
Row swap	$\binom{10}{2} = 45$	43	No improvement
Column swap	$\binom{10}{2} = 45$	42	No improvement
Symbol relabel	$\binom{10}{2} = 45$	26	No improvement
Single intercalate	≤ 2500	28	No improvement

Theorem 3.1 ($\mathcal{E} = 26$ is a strict local minimum). *The $\mathcal{E} = 26$ triple (L_1, L_2, L_3) for pair `iso_tv_21_0` is a strict local minimum under all standard Latin square moves: every row swap, column swap, symbol relabeling, and single intercalate flip on L_3 yields $\mathcal{E} \geq 26$, with strict inequality for all moves except symbol relabelings that preserve $\mathcal{E} = 26$.*

3.5.2 Double-Intercalate Analysis

All 281 pairs of distinct intercalate positions were tested as double-intercalate flips:

Theorem 3.2 (Double-intercalate optimality). *For the $\mathcal{E} = 26$ solution, every one of the 281 double-intercalate combinations yields $\mathcal{E} \geq 26$. That is, no pair of simultaneous intercalate flips on L_3 can reduce the energy below 26.*

3.5.3 4-Row LNS Analysis

All $\binom{10}{4} = 210$ subsets of 4 rows were freed and re-optimized by CP-SAT, with the remaining 6 rows fixed:

Theorem 3.3 (4-row LNS local optimality). *For every 4-row subset of the $\mathcal{E} = 26$ solution, freeing those 4 rows and re-optimizing with CP-SAT returns $\mathcal{E} = 26$ trivially in under 0.1 seconds. The $\mathcal{E} = 26$ solution is 4-row locally optimal.*

3.5.4 Unique-Basin Property

Theorem 3.4 (Unique attractor basin). *Five independent CP-SAT trials (each 30 seconds, different random seeds), all initialized with the $\mathcal{E} = 26$ hint L_3 , all return the same canonical L_3 matrix. No trial found a different L_3 with $\mathcal{E} \leq 26$.*

This unique-attractor property strongly suggests the $\mathcal{E} = 26$ state sits in a deep, narrow basin with no competing states at the same energy.

Table 8: Feasibility attack: 6 trials targeting $\mathcal{E} \leq 25$ for iso_tv_21_0.

Trial	Timeout (s)	Workers	Result	Cumulative worker-seconds
1	600	2	TIMEOUT	1,200
2	600	2	TIMEOUT	2,400
3	600	2	TIMEOUT	3,600
4	600	2	TIMEOUT	4,800
5	600	2	TIMEOUT	6,000
6	600	2	TIMEOUT	7,200
Total	—	—	0 SAT, 0 INFEASIBLE	7,200

3.5.5 Feasibility Attack: $\mathcal{E} \leq 25$

Agents launched a direct feasibility attack with the hard constraint $\mathcal{E} \leq 25$:

All 6 trials timed out. Neither SAT (finding L_3 with $\mathcal{E} \leq 25$) nor INFEASIBLE (proving no such L_3 exists) was returned. The 7,200 worker-seconds of compute without any SAT result provides strong empirical evidence that $\mathcal{E} = 26$ is the global minimum for this pair, but a conclusive proof remains open.

3.5.6 Secondary Cascade: A Second L_3 at $\mathcal{E} = 28$

Using a different random seed on the same pair **ee29e34e**, agents discovered a secondary cascade reaching $\mathcal{E} = 28$ ($\text{cl}_{13} = 16$, $\text{cl}_{23} = 12$, seed=2085968723). Canonical-form analysis confirmed this is a *genuinely distinct* L_3 from the $\mathcal{E} = 26$ solution (different matrix). This secondary cascade used pool $\mathcal{E} = 29$ entries as hints.

3.6 Stage 5 (Sessions 29–30): Multi-Pair Assault, Three Independent Pairs at $\mathcal{E} = 28$

3.6.1 mdecomp_279 Reaches $\mathcal{E} = 28$

Following the $\mathcal{E} = 26$ breakthrough, agents pivoted to other high-potential pairs. The mdecomp_279 cascade (fingerprint **e54e791c**, a completely different L_1 family):

- Cold trial (Session 28): $\mathcal{E} = 29$ ($\text{cl}_{13} = 12$, $\text{cl}_{23} = 17$)—dramatic improvement over the scan record of $\mathcal{E} = 33$ for this pair.
- Session 29, trial 4 (hint=E29): $\mathcal{E} = \mathbf{28}$ ($\text{cl}_{13} = 8$, $\text{cl}_{23} = 20$, seed=1046158808). A completely independent (L_1, L_2) pair matches the secondary cascade record.

3.6.2 Systematic Scan of 115 New mdecomp Pairs

Agents catalogued and scanned 115 previously unexplored mdecomp CT=2 pairs (120 seconds per trial). Key findings from the scan:

- mdecomp_56 (cold scan): $\mathcal{E} = 31$ in 120 seconds.
- mdecomp_113 (cold scan): $\mathcal{E} = 31$ in 120 seconds.
- Discovery: mdecomp_56 $\equiv$ mdecomp_68 (same L_1, L_2 hash fingerprints despite different decomposition indices).
- mdecomp_56 cascade, trial 2: $\mathcal{E} = \mathbf{30}$ ($\text{cl}_{13} = 15$, $\text{cl}_{23} = 15$, seed=1647156375). Notably balanced: both orthogonality deficits equal.

3.6.3 mdecomp_113 Reaches $\mathcal{E} = 28$: The Third Independent Pair

The mdecomp_113 cascade (L_1 fingerprint 2469c6dc):

- Trial 1–2: $\mathcal{E} = 31$ (cold start, confirming scan result).
- Trial 3 (hint=E31): $\mathcal{E} = 28$ ($cl_{13} = 17$, $cl_{23} = 11$, seed=1296853556). A *third structurally independent* MOLS-10 pair reaches $\mathcal{E} = 28$.
- Trials 4–5 (hint=E28): $\mathcal{E} = 28$ (cascade stable at $\mathcal{E} = 28$; searching for $\mathcal{E} < 28$).

3.6.4 Three Independent Pairs at $\mathcal{E} = 28$: Summary

Theorem 3.5 (Three independent pairs at $\mathcal{E} = 28$). *Three structurally distinct MOLS-10 pairs, from different L_1 families, all reach $\mathcal{E} = 28$ via CP-SAT hint cascade optimization:*

Table 9: Three independent MOLS-10 pairs converging to $\mathcal{E} = 28$.

Pair	L_1 fingerprint	cl_{13}	cl_{23}	$\mathcal{E}$	Seed
iso_tv_21_0 (secondary)	ee29e34e	16	12	28	2085968723
mdecomp_279	e54e791c	8	20	28	1046158808
mdecomp_113	2469c6dc	17	11	28	1296853556

The convergence of three structurally independent pairs to the same energy level $\mathcal{E} = 28$ strongly suggests this is a structural feature of the energy landscape of MOLS-10, not a coincidence.

3.7 Stage 6 (Sessions 31+): Scaled Parallel Search and Landscape Survey

3.7.1 Fast Incremental SA with $O(1)$ Energy Updates

Sessions 31+ developed a significantly faster SA implementation using pair-frequency tables. Instead of recomputing $cl_{13} + cl_{23}$ from scratch ($O(N^2)$) after each intercalate move, incremental updates maintain the $N \times N$ frequency matrices $f_{13}[a][b] = |\{(r, c) : L_1[r, c] = a, L_3[r, c] = b\}|$ and f_{23} analogously. Each intercalate flip affects exactly 8 entries across the two tables. The energy is then the count of zero entries: $\mathcal{E} = |\{(a, b) : f_{13}[a][b] = 0\}| + |\{(a, b) : f_{23}[a][b] = 0\}|$.

This achieves approximately 5,000–6,000 steps/s compared to 1,000–1,500 steps/s in previous SA implementations. Continuous restarts from the $\mathcal{E} = 26$ anchor confirm that it is a strict attractor: every restart from a perturbed anchor returns to $\mathcal{E} \geq 26$, and the best random-start run from scratch reaches only $\mathcal{E} = 52$.

3.7.2 Landscape Universality: All Tested Minima are Strict 2-Deep IC Local Minima

BFS analysis was extended from iso_tv_21_0 to additional pairs:

- **mdecomp_113** ($\mathcal{E} = 28$): BFS depth 5, threshold $\mathcal{E} \leq 40$, found 15,524 unique states with global minimum $\mathcal{E} = 28$. The minimum has 33 intercalates and all 1-IC neighbors have $\mathcal{E} \geq 30$.
- **mdecomp_279** ($\mathcal{E} = 28$): BFS found 20 intercalates; all 1-IC neighbors have $\mathcal{E} \geq 30$ (strict 2-deep minimum).
- **mdecomp_56** ($\mathcal{E} = 30$): has a flat 1-IC neighbor at $\mathcal{E} = 30$ (non-strict), but no 1-IC neighbor with $\mathcal{E} < 30$.

The universal pattern across all tested pairs: the best known energy state is either a strict local minimum or a plateau under intercalate moves. No pair has shown a gradient pointing toward lower energy that SA can follow.

3.7.3 Systematic Survey of Unscanned Pairs

A survey of previously unscanned pairs was launched using 200k-step random-start SA:

- **76 unscanned mdecomp pairs:** Initial survey results show $\mathcal{E} \in [53, 65]$ from random starts (expected for short random-start SA). No pair yet shows scan energy below $\mathcal{E} = 53$ from quick trials. Deep search of top candidates continues.
- **193 non-mdecomp pairs** (tv, iso, sa.ct, ic variants): These pairs—including 103 sa.ct pairs generated specifically to maximize CT—had never been searched for a compatible L_3 . A cross-seeded survey uses known best L_3 solutions (from $\mathcal{E} = 26$ and $\mathcal{E} = 28$ anchors) as starting points to quickly characterize each pair.

All 103 sa.ct pairs have CT = 2 (confirmed). The sa.ct_climb algorithm searched for higher CT without success; CT = 2 appears to be a universal ceiling for MOLS-10.

3.7.4 New Search Strategies Evaluated

Parallel Tempering SA. A 4-replica parallel tempering SA ($T \in \{15, 3, 0.5, 0.05\}$) was implemented and evaluated. The fundamental failure mode: the energy gap between high-temperature replicas ($\mathcal{E} \approx 70\text{--}80$) and low-temperature replicas ($\mathcal{E} = 26$) is too large for productive swap moves. Accept rates for low-T replicas were 0%, and no swaps involving the cold replica were accepted. This confirms the deep isolation of the $\mathcal{E} = 26$ basin.

Genetic Algorithm with Latin Square Repair. A population-based GA with row-crossover and iterative repair to the Latin square constraint was implemented and fixed. (Initial versions had a latent bug in the repair routine producing invalid Latin squares; this was corrected by adding an iterative repair with validity checking.) The GA converges to $\mathcal{E} = 26$ but does not improve beyond it.

2-Row Exhaustive Optimizer. A hybrid approach alternates between standard SA intercalate moves and an exhaustive optimizer: every 50k SA steps, two rows are temporarily removed from L_3 ; the optimal assignment for those two rows is found by backtracking over all valid ($\text{row}_1, \text{row}_2$) pairs (at most $2^{10} = 1024$ combinations) and the best is installed. This allows moves equivalent to multiple simultaneous intercalates. Evaluation ongoing.

4 Summary and Discussion

4.1 Summary of Main Results

Table 10 compiles the main quantitative results of the 31+ session search.

4.2 The CT Barrier and the $N(10)=2$ Conjecture

The most striking structural observation from this project is the universality of the CT barrier: *Conjecture 4.1* (CT Barrier Conjecture). Every MOLS-10 pair (L_1, L_2) satisfies $\text{CT}(L_1, L_2) \leq 2$.

If Conjecture 4.1 is true, then by Corollary 1.7, $N(10) = 2$ —settling the 60-year-old open problem. Our evidence for this conjecture is strong but not conclusive:

- All 344 pairs tested (from 28 distinct L_1 families, using 5 different construction methods) have $\text{CT} \leq 2$.
- The SA-CT climbing algorithm actively seeks pairs with high CT, yet never exceeds CT=2 despite hundreds of trials.
- Glucose3 provides machine-verifiable UNSAT proofs for all 323 pairs.

Table 10: Summary of main results from 31+ sessions of autonomous search.

Result	Details
All-time energy record	$\mathcal{E} = 26$ ($cl_{13} = 8$, $cl_{23} = 18$), pair ee29e34e , seed=1943590465
Energy cascade	$37 \rightarrow 34 \rightarrow 33 \rightarrow 32 \rightarrow 31 \rightarrow 30 \rightarrow 29 \rightarrow 26$ (Sessions 5–26b)
E=26 local min	Strict under all row/col/symbol/intercalate 1-moves
E=26 2-IC analysis	All 281 double-intercalate combos yield $\mathcal{E} \geq 26$
E=26 LNS	All 210 four-row subsets return $\mathcal{E} = 26$ in < 0.1 s
E=26 unique basin	5 independent CP-SAT trials converge to same L_3
E=26 BFS depth-6	77,507 unique states explored; global min = 26
Feasibility attack	7,200 worker-seconds with $\mathcal{E} \leq 25$ constraint: 0 SAT, 0 INFEASIBLE
Three pairs at E=28 mdecomp_113 BFS	ee29e34e (secondary), e54e791c , 2469c6dc 15,524 states at depth 5; global min = 28 (strict 2-deep)
Pairs at E=29	4 isotopy classes; BFS depth-5 global IC-min proved
Fast SA speed	5,000–6,000 steps/s via incremental frequency-table updates
MOLS-10 pair catalog	323 pairs (Stage 1–3) + 191 new pairs surveyed = 514+ pairs tested
CT barrier	All 514+ pairs: $CT(L_1, L_2) \leq 2 < 10$
SAT certificates	323 UNSAT certificates (Glucose3, ≈ 42 ms each)
L_1 families covered	28+ distinct L_1 families across 323 pairs

The total space of MOLS-10 pairs is astronomically large; exhaustive verification is computationally infeasible. A mathematical proof of the CT barrier would resolve the problem definitively.

4.3 The Energy Landscape: Structural Observations

The experimental data reveals a rich structure in the energy landscape:

The SA floor ($\mathcal{E} = 37$). SA cannot escape $\mathcal{E} = 37$ because it is a strict local minimum under all standard moves. Breaking this barrier required a non-local optimization method (CP-SAT). The floor is remarkably robust: it persisted across all pairs, all pool sizes, and all SA hyperparameter settings.

The E=28 platform. Three structurally independent pairs (L_1 families **ee29e34e**, **e54e791c**, and **2469c6dc**) all converge to $\mathcal{E} = 28$ via hint cascade. This suggests a structural “platform” in the energy landscape at $\mathcal{E} = 28$: a region where many distinct local optima cluster. Whether $\mathcal{E} = 28$ represents a fundamental combinatorial barrier or merely reflects current algorithm limitations is unknown.

The E=26 basin. The $\mathcal{E} = 26$ state is the deepest known point in the energy landscape. Its properties (strict local minimum, unique attractor basin, 7,200 worker-seconds of failed improvement attempts) suggest it is either the global minimum for pair **ee29e34e** or a very deep

local minimum with an extremely small basin of improvement. The 3-unit jump from $\mathcal{E} = 29$ to $\mathcal{E} = 26$ in a single CP-SAT trial shows the search can make large leaps; a comparable leap from $\mathcal{E} = 26$ to $\mathcal{E} = 0$ would solve $N(10)$.

4.4 Evidence Concerning $N(10)=2$

Combining all results, the evidence supporting $N(10) = 2$ includes:

1. **Universal CT barrier.** All 514+ tested pairs have $CT \leq 2$, with machine-verifiable proofs for all 323 of the original pairs.
2. **Deep energy minimum.** $\mathcal{E} = 26$ is a strict local minimum with a unique attractor basin; BFS depth-6 confirms 77,507 intercalate-reachable states all have $\mathcal{E} \geq 26$. The best random-start SA from scratch reaches only $\mathcal{E} = 52$.
3. **Landscape universality.** All tested energy minima (iso_tv_21_0 at $\mathcal{E} = 26$; mdecomp_113, mdecomp_279 at $\mathcal{E} = 28$; mdecomp_56 at $\mathcal{E} = 30$) are strict or near-strict 2-deep local minima. No gradient toward lower energy is found anywhere.
4. **Convergence barrier.** Three independent pairs stall at $\mathcal{E} = 28$; the pairs at $\mathcal{E} = 29$ are BFS-depth-5 global intercalate-minima.
5. **Exhaustive feasibility failure.** 7,200 worker-seconds of CP-SAT with hard constraint $\mathcal{E} \leq 25$ found neither a solution nor a proof of infeasibility, suggesting $\mathcal{E} = 25$ is very hard to reach.
6. **Scale of exploration.** 28 distinct L_1 families, 6+ generation methods, 31+ research sessions, all telling the same story.

None of this constitutes a mathematical proof; the question remains open.

4.5 The Autoresearch Framework: Lessons Learned

The autonomous agentic approach proved essential for this investigation:

- A single human researcher could not have sustained 30 sessions of systematic pair scanning, cascade management, and algorithm iteration at this pace.
- Key breakthroughs (the hint cascade strategy, the BFS local-minimum proof, the feasibility attack design) were *invented* by agents responding to experimental data, not pre-programmed.
- The agents' ability to run hundreds of parallel trials, kill stalled processes, and immediately redirect compute to breakthroughs was critical to achieving $\mathcal{E} = 26$.

The framework demonstrates that autonomous AI agents can conduct genuinely productive open-ended mathematical research, including algorithm design, implementation, and strategic adaptation.

4.6 Open Questions and Future Directions

1. **Prove or disprove the CT Barrier Conjecture.** A mathematical proof that all MOLS-10 pairs have $CT \leq 2$ would resolve $N(10) = 2$. Alternatively, finding a MOLS-10 pair with $CT \geq 10$ would be a major breakthrough toward $N(10) \geq 3$.
2. **Improve the energy record.** Can $\mathcal{E} < 26$ be achieved? CP-SAT with longer timeouts, distributed search, or new algorithmic ideas might escape the current basin.

3. **Explain the E=28 platform.** Why do three independent pairs converge to exactly $\mathcal{E} = 28$? Is there a combinatorial invariant of MOLS-10 pairs that implies $\mathcal{E} \geq 28$ for all pairs and all L_3 ?
4. **Extend the pair catalog.** Are there MOLS-10 pairs with structural properties fundamentally different from those in the current 323+115-pair catalog? In particular, are there pairs with $\text{CT} > 2$?
5. **Prove a lower bound on $\mathcal{E}$.** A theoretical argument showing $\mathcal{E} \geq c$ for some $c > 0$ and all MOLS-10 pairs would give a conditional proof of $N(10) = 2$ given the CT barrier.
6. **Scale up computation.** Distributed CP-SAT across hundreds of CPUs, or GPU-accelerated search, might find $\mathcal{E} = 0$ or provide tighter feasibility bounds.

4.7 Conclusion

We have presented the most thorough computational investigation of the $N(10)$ problem to date, enabled by an autonomous AI agentic research framework operating over 31+ sessions. The energy $\mathcal{E} = \text{cl}_{13} + \text{cl}_{23}$ was driven from the SA floor of 37 to an all-time record of 26, passing through the levels 34, 33, 32, 31, 30, 29 along the way.

The record $\mathcal{E} = 26$ is a strict local minimum with a unique attractor basin, BFS-verified to depth 6 across 77,507 unique intercalate-reachable states (all with $\mathcal{E} \geq 26$). Three independent MOLS-10 pairs from structurally distinct L_1 families all stall at $\mathcal{E} = 28$, and BFS analysis of `mdecomp_113` ($\mathcal{E} = 28$) also yields a strict 2-deep local minimum. Every one of the 514+ pairs in our catalog satisfies the CT barrier ($\text{CT} \leq 2$), with machine-verifiable Glucose3 proofs for all 323 of the original pairs.

Collectively, these findings provide the strongest computational evidence yet that $N(10) = 2$. A mathematical proof of the CT barrier conjecture would settle the problem definitively.

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